

Jonathan Jacob Koshy

Portfolio | johnkoper12@gmail.com | LinkedIn | GitHub: JJKSweaty

Education

University of Waterloo
BASc. Electrical Engineering

Waterloo, ON
Expected 2029

Experience

Embedded Software Intern – Belimo - Montreal, QC May 2026 – Aug 2026

- Developed bare metal **Embedded C** on Advanced 8 bit **PIC18 MCU**, reducing memory use by **20%**
- Validated **CAN, UART, and I2C** signals with lab tools to isolate timing faults and controller state bugs
- Optimized interrupt timing and debounce windows, improving keypad navigation latency by **30%**
- Implemented **ADC digital filtering** for gas sensing firmware, improving measurement accuracy by **10%**
- Improved **ModBus** communication by debugging UART parsing, frame handling, and state recovery faults
- Tested firmware against UL verification guidelines across sensor behavior, edge cases, and PCB level validation

Embedded Software Intern – AeroCardia - Montreal, QC Sep 2025 – Dec 2025

- Developed **C++ sensor** drivers for IMU, PPG, and temperature sensors, enabling **50 Hz** biosignal capture
- Built an **RTOS** queue pipeline decoupling sensor acquisition from **BLE** transmission for deterministic scheduling
- Optimized sensor sampling and buffering paths, reducing dropped samples by **25%** during continuous monitoring
- Implemented secure **BLE OTA** updates with image verification and rollback, cutting update time to **under 1 min**
- Designed an O2 sensor PCB in **Altium** with onboard EEPROM for persistent sensor calibration data

Embedded Software Member – UWARG – Waterloo, ON Dec 2024 – Sep 2025

- Implemented roll and yaw motor mixing firmware, mapping pilot inputs to motor outputs in fixed-wing modes
- Tuned **PID** attitude loops, reducing manual flight oscillation by **20%** during stabilization tests
- Improved control loop timing and actuator output scaling, reducing motor command jitter by **30%**

Embedded Software Member – Electrium Mobility – Waterloo, ON May 2025 – Aug 2025

- Built **ESP32** firmware for an electric skateboard dashboard, streaming **BLE** telemetry using FreeRTOS
- Integrated **VESC** motor controller data over **UART**, streaming RPM, current, and battery metrics
- Added packet parsing and checksum validation for skateboard diagnostics, reducing malformed frames by **25%**

Projects

Custom VR/AR Headset C, STM32F411, ICM20948, TF-Luna LiDAR, USB HID, FastIMU, KiCad 

- Built custom 2K 120Hz VR headset with 9DoF orientation, LiDAR depth, and SteamVR integration
- Developed STM32F411 firmware to sample and scale ICM20948 accel, gyro, and magnetometer data over I2C
- Implemented FastIMU quaternion fusion and USB HID reports to stream orientation data into SteamVR
- Integrated TF-Luna LiDAR ranging to measure forward displacement for seated positional tracking
- Designed 2-layer KiCad PCB with USB-C, fused 5V display power, 3.3V regulation, and test points

Edge AI Dementia Companion (HackCanada Winner) C, Python, YOLOv8, Hailo-8L, PCA9685, RPi 5 

- Built edge AI memory aid on RPi 5 and Hailo-8L that locates misplaced objects for dementia patients
- Accelerated YOLOv8s inference at 1080p 30fps, queuing detections for persistent last-seen object memory
- Drove dual-axis pan-tilt servos through PCA9685 PWM to sweep rooms and recover objects outside camera view

ESP32 Wi-Fi Media Controller C++, Python, ESP32, LVGL, TCP/IP, WebSockets  

- Built a custom handheld media dashboard with **LVGL** for PC metrics, media state, and Discord data
- Designed a TCP JSON protocol with queued updates, RGB565 artwork rendering, and touchscreen media controls

Skills

- **Programming:** C, C++, Python, RISC-V, MATLAB, Git
- **Protocols/Platforms:** SPI, I2C, UART, BLE, CAN, Ethernet, PWM, RTOS, ARM Cortex, Raspberry Pi, Linux
- **Lab Tools:** Oscilloscopes, Logic Analyzers, Multimeters, Soldering Irons, Bench Power Supplies